

Vatsal Verma

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PROFESSIONAL SUMMARY

Backend Software Engineer with nearly 3 years of experience building scalable microservices and distributed systems in Java, Spring Boot, and Spring Cloud. Strong in event-driven architecture (Apache Kafka), REST APIs (OpenAPI, OAuth2/JWT), and cloud-native deployment (Docker, Kubernetes, OpenShift). Delivered production GenAI capabilities including an LLM-powered agent with a RAG pipeline using hybrid search retrieval (BM25 + embeddings) and MCP tool integrations for production incident triage and remediation. Track record in CI/CD optimization, application security (CVE/OWASP), and end-to-end feature ownership for high-scale telecom billing platforms.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Backend & Distributed Systems: Spring Boot, Spring Cloud, Spring Security, microservices architecture, REST APIs, OpenAPI, OAuth2/JWT, system design, OOP, data structures and algorithms

Data, Search & Messaging: Apache Kafka, Couchbase, Elasticsearch, SQL, event-driven architecture

AI / GenAI: Retrieval-Augmented Generation (RAG), LLM-powered AI agents, GenAI, hybrid retrieval (BM25 + embeddings), semantic search, vector database, MCP (Model Context Protocol), Google ADK, NLP; TensorFlow, Keras

DevOps & Testing: Docker, Kubernetes, Red Hat OpenShift, Jenkins, Maven, Git, CI/CD, JUnit, TestNG

PROFESSIONAL EXPERIENCE

Software Developer Experienced (*Software Engineer - Java & AI*)

August 2023 – Present

Amdocs, Pune, India

- Developed and delivered key features within a high-scale billing ecosystem of 36 microservices. Contributed to end-to-end feature lifecycles in Agile/Scrum sprints for major Tier-1 telecom providers, including AT&T and Bell Canada, serving 50+ million subscribers
- Developed event-driven architecture and REST API microservices with Java, Spring Boot, Spring Cloud, Apache Kafka, and Couchbase; implemented OpenAPI, OAuth2/JWT security, and API rate limiting
- Built production LLM-powered agent for incident remediation with a Retrieval-Augmented Generation (RAG) pipeline using hybrid retrieval (BM25 lexical + dense vector embedding search) as the retrieval engine and Model Context Protocol (MCP) tool integrations; used Elasticsearch as the vector database (**dense_vector** fields with kNN search) to store the RAG knowledge base, embeddings, and prior root cause analyses (RCAs); enriched context with Apache Kafka events and REST APIs; reduced root cause analysis time by 20% and cut manual incident ticket handling
- Optimized Maven-based CI/CD on OpenShift, reducing integration test time by 25%
- Resolved 42 CVE and OWASP issues using GenAI-assisted code review (Cursor, Copilot); owned secure production releases

AI/ML Research Intern

August 2021 – February 2022

Indian Institute of Technology (IIT-BHU), Varanasi, India

- Designed facial recognition with Convolutional Neural Networks (CNNs), TensorFlow, and Keras for identification of 50,000+ individuals
- Extended models for agricultural applications by analyzing unique muzzle patterns of 100+ cattle using computer vision techniques
- Researched deep learning and image processing pipelines; managed timelines and presented findings to faculty advisors

PROJECTS

AI-Powered Animal Detection and Alert System

Python, TensorFlow, Keras, OpenCV, REST API

- Built an end-to-end deep learning inference pipeline with CNNs (TensorFlow/Keras) and OpenCV for real-time animal detection (92.68% accuracy); integrated preprocessing and data augmentation for robustness across lighting conditions
- Delivered low-latency event notifications via Twilio REST API; recognized at WildHacks for creative API integration

NLP Document Summarization Service

Java, Spring Boot, Spring REST, Maven, NLP

- Developed a Spring Boot REST microservice with an NLP pipeline (tokenization, stopword removal, sentence segmentation) and LexRank extractive summarization; validated output against reference summaries (85+% accuracy)
- Designed modular ingestion and ranking stages exposed via HTTP APIs, using standard Java microservice patterns (layered controllers, service layer, and REST DTOs)

EDUCATION

Vellore Institute of Technology, Vellore

July 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering with specialization in IoT

CGPA: 8.22/10

ACHIEVEMENTS AND LEADERSHIP

- Innovation Award, Amdocs:** Recognized for enhancing TestNG suite execution and CI/CD efficiency using Kubernetes APIs for cloud cluster integration
- WildHacks 2021:** Won prize for most creative use of Twilio API in hackathon competition
- Design Head, ISA-VIT Student Chapter:** Managed a team of 15; designed and released an application on Google Play Store